NATIONAL ECONOMICS UNIVERSITY
FACULTY OF DATA SCIENCE AND ARTIFICIAL INTELLIGENCE

GROUP ASSIGNMENT REPORT

Midterm

Web Design and Programming (Course code)

Your Project Title

One-line description of the product

Group: Group 01
Class: Class name
Instructor: M.Sc. Le Duy Khanh
Semester: Semester 1, 2026–2027
GitHub: https://github.com/your-group/your-repo
Demo: https://your-group.github.io/your-repo

Team Members

No.	Full name	Student ID	Role
1	Nguyen Van A	11xxxxxxxx	Team lead, Back-end
2	Tran Thi B	11xxxxxxxx	Front-end, UI/UX
3	Le Van C	11xxxxxxxx	Database, testing
4	Pham Thi D	11xxxxxxxx	Documentation, deployment

Hanoi, November 2026

Declaration

We declare that this report is the result of our group's own work for the course *course*. All sources (literature, source code, images, AI tools) are cited explicitly. We take full responsibility for the submitted content.

Task Allocation and Contributions

Member	Work done	Contribution (%)
Nguyen Van A	API design, authentication, repository management	25
Tran Thi B	Interface design, responsive layout, CSS/JS	25
Le Van C	Database design, tests, integration testing	25
Pham Thi D	Report writing, deployment, demo video	25
	Total	**100**

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Contributions must sum to 100%. Base them on the commit history and your meeting notes. The instructor may use them to adjust individual grades.

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1. Introduction

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About 1–2 pages. Answer: what did the group build, for whom, and why does it matter?

1.1. Background and motivation

Describe the real-world context, the problem to solve, and why your group chose this topic.

1.2. Objectives

- Objective 1 (specific and measurable).
- Objective 2.
- Objective 3.

1.3. Scope and target users

Who is the product for? Which features are *in scope* and which are *out of scope*?

1.4. Technologies

Component	Technology	Purpose
Front-end	HTML5, CSS3, JavaScript	User interface
Back-end	Node.js / Express	Business logic, API
Database	MongoDB / PostgreSQL	Data storage
Tooling	Git, GitHub, Figma	Version control, design

Table 1. Technologies used in the project.

1.5. Report structure

Section 2 covers analysis and design; Section 3 describes the implementation; Section 4 presents results and conclusions.

2. Analysis and Design

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Explain your *design decisions* and the reasons behind them, not just the results. Figures must remain readable when printed in black and white.

2.1. Requirements

2.1.1. Functional requirements

FR1. Users can register and log in.

FR2. …

2.1.2. Non-functional requirements

Performance, security, accessibility, responsiveness on mobile devices…

2.2. Interface design (UI/UX)

Insert wireframes or mockups (e.g. from Figma). Explain the layout, colour palette and typography.



Figure 1. Placeholder figure — replace with the home page wireframe.

2.3. System architecture

Describe the client–server model, the request/response flow and the main components.

2.4. Database design

Provide the ER diagram and describe the main tables or collections.

2.5. API design

Method	Endpoint	Description
GET	`/api/items`	List items
POST	`/api/items`	Create an item
PUT	`/api/items/:id`	Update an item

Table 2. Main API endpoints.

3. Implementation

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Quote only the *key* code snippets (under 25 lines each) and explain them. Keep the full source on GitHub; do not paste whole files into the report.

3.1. Project structure

```
project/
|-- public/          # static assets
|-- src/
|   |-- routes/
|   |-- models/
|   `-- app.js
|-- package.json
`-- README.md
```

3.2. User interface

Include screenshots of the main pages (desktop and mobile) with short descriptions.

3.3. Server-side logic

Listing 1. Example login route.

```
app.post('/api/login', async (req, res) => {
  const { email, password } = req.body;
  const user = await User.findOne({ email });
  if (!user || !(await bcrypt.compare(password, user.hash))) {
    return res.status(401).json({ error: 'Invalid credentials' });
  }
  res.json({ token: sign(user) });
});
```

Listing 1 …

3.4. Source control and teamwork

Describe your Git workflow (branches, pull requests, code review). Link the repository here (see the cover page).

3.5. Deployment

State the hosting platform, configuration and the public URL.

4. Results and Conclusion

4.1. Testing

ID	Feature	Expected result	Result
TC01	Valid login	Redirect to home page	Pass
TC02	Invalid login	Show an error message	Pass
TC03	Responsive 375px	Layout does not break	Pass

Table 3. Selected test cases.

Add Lighthouse scores (Performance, Accessibility, Best Practices, SEO) with a screenshot.

4.2. Achievements and limitations

Compare against the objectives in Section 1: which were met, and to what extent? State honestly the known limitations and remaining bugs.

4.3. Lessons learned and future work

Difficulties encountered, how the group solved them, and what you would do differently.

- Possible improvement 1.
- Possible improvement 2.

References

[1] Jon Duckett. *HTML and CSS: Design and Build Websites*. Wiley, 2011.

[2] Mozilla. Mdn web docs, 2026. Accessed: 2026-11-01.

A. Appendix

A.1. Setup and run instructions

```
git clone https://github.com/your-group/your-repo.git
cd your-repo
npm install
npm start
```

A.2. Demo accounts

Role	Account / Password
Admin	admin@example.com / …
User	user@example.com / …

A.3. Use of AI tools (if any)

List the AI tools used, for what purpose, and which parts of the product they helped with.